

DEPARTMENT OF COMPUTER SCIENCE (CYBERSECURITY)

DEEPFAKES AND AI-POWERED CYBER THREATS

ABSTRACT

The rapid development of Artificial Intelligence has introduced new opportunities in cybersecurity, but it has also created sophisticated methods for conducting cyberattacks. One emerging threat is the use of AI-generated content, particularly deepfakes, to manipulate images, videos, audio, and text. Attackers can use these technologies to impersonate individuals, create convincing fraudulent communications, spread misinformation, or perform highly targeted social engineering attacks. This seminar explores how generative AI is changing the cybersecurity threat landscape and how deepfake technology can be exploited to deceive individuals and organizations. It focuses on threats such as AI-generated phishing messages, voice impersonation, fake video communications, automated social engineering, and identity fraud. The seminar also examines the difficulties involved in detecting AI-generated content, as modern generation techniques can produce increasingly realistic results. Various defensive approaches, including digital watermarking, content authentication, multi-factor verification, AI-based detection systems, and employee awareness training, are discussed. The topic highlights the importance of combining technological solutions with human verification practices to reduce the risks associated with AI-powered deception. Understanding these emerging threats is essential for developing effective cybersecurity strategies as artificial intelligence continues to become more accessible and powerful.

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